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Joan PASCUAL RIBES, 19 years old. Spanish Nationality

JPR

*Second year Mathematics and Engineering Physics student. Polymath.  
Passionate about knowledge. Hungry learner.*

## Education

**2023 - December 2027** Degree in **Engineering Physics**, Polytechnic University of Catalonia.  
Degree in **Mathematics**, Polytechnic University of Catalonia.  
CFIS (Interdisciplinary Higher Education Centre)

## Skills

**Programming** C++, Intermediate.  
Python, Intermediate.  
LaTeX, Intermediate.

**Languages** English (Advanced) C1 certificate, *Trinity College London*.  
Spanish (Native).  
Catalan / Valencian (Native). C2 certificate.  
German (Basic) Currently studying B1 level.  
French (Basic) A2 certificate.

**Teamwork and Communication** I enjoy working in pluralistic teams and learning from others. In collaborative research and programming projects, I have communicated effectively and contributed to achieving objectives.

## Academic Achievements

**Double Degree Mark** 9.33 / 10.

**Honours** (*Mathematics*): One Variable Calculus, Linear Algebra, Integral Calculus, Multilinear Algebra and Projective Geometry.  
(*Engineering Physics*): Physics I, Physics II, Biophysics, Thermodynamics and Quantum Physics.

**PAU Mark** 13.832 / 14.

**High School Mark** 10 / 10. (Honours)

## Work Experience

**Barcelona Introduction to Mathematical Research** *July, 2024.* At the CRM (Centre de Recerca Matemàtica), in a collaboration between the universities UAB, UB and UPC. A month of research in the field of Algebra. We studied polynomials and their expression as sum of squares.

## Main Interests

**Competitive Programming** I enjoy solving hard problems while learning fast algorithms and problem-solving strategies.

**Trading and Markets** The connection it has to the fascinating academic fields of probability, statistics, and game theory particularly appeals to me.

**Computer Science** I am truly interested in AI, computer vision and machine learning, specially when it comes to their relation with Mathematics.

## Olympiad Experience

|                         |                                                                  |
|-------------------------|------------------------------------------------------------------|
| <b>Mathematics</b>      | Winner of Cangur Olympiad (Valencia) (2023).                     |
| <b>Physics</b>          | Autonomic Physics Olympiad (Valencia). (Third place) (2023).     |
| <b>Chemistry</b>        | Participation in the Spanish National Chemistry Olympiad (2023). |
| <b>Catalan Language</b> | Winner of two consecutive autonomic contests (2022 & 2023).      |

## Projects

|                                       |                                                                                                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ecosystem Simulation and Study</b> | We have developed a magnificent tool for understanding and simulating complex ecosystem dynamics. In addition, we have also conducted a study on the results presented by the simulation. <a href="#">Link to the project</a> .                 |
| <b>Simulation of Superfluidity</b>    | After a thorough investigation of the physical phenomenon of superfluidity, we developed a series of simulations that, using spectral methods, replicate the characteristics of the Helium-II superfluid. <a href="#">Link to the project</a> . |

## Hobbies and Activities

|                         |                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------|
| <b>Literature</b>       | Awarded in some literature contests. I genuinely enjoy poetry and philosophical essays.  |
| <b>Music</b>            | I have played in a traditional band and studied saxophone and piano at the conservatory. |
| <b>Webpage creation</b> | Currently learning the basics of HTML and starting <a href="#">my personal website</a> . |

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The most important thing for me is to gain experience and learn new concepts while facing challenging problems.